Knowledge Unlearning using Sparse AutoEncoders

TLDR; We want LLM’s to forget Harry Potter (& wizarding universe)

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Defining the Problem

IP (Intellectual Property) Memorization in LLMs

Large Language Models (LLMs) often inadvertently memorize copyrighted intellectual property (IP) during pre-training, leading to ethical and legal risks.

Traditional unlearning methods rely on weight modification, which lacks transparency and risk "catastrophic forgetting" of general knowledge.

Background & Existing Methods

Approximate Unlearning

Eldan & Russinovich (2023) proposed fine-tuning to forget content by replacing idiosyncratic terms with generic alternatives.

1. Detect Harry Potter related tokens
2. Compare logits with baseline model
3. Replace specific phrases with generic ones
4. Generate alternative next token predictions
5. Finetune model using new labels

Limitations

- Unlearning may affect global behavior
- May forget relevant non-target information
- Hard to isolate exact knowledge region

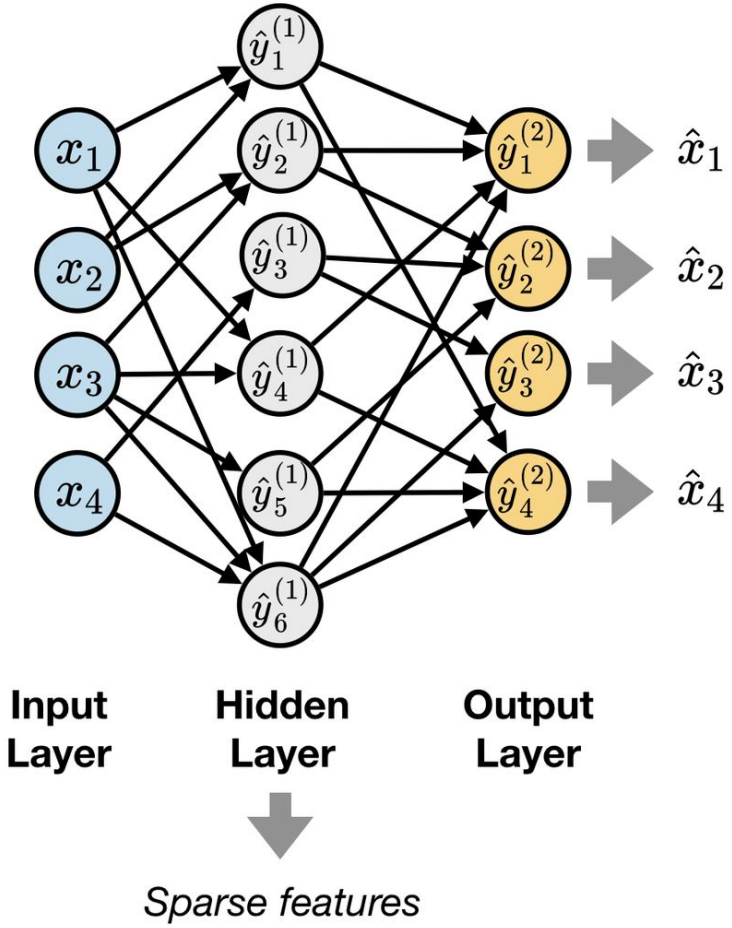
Sparse Autoencoders (SAE)

What is an SAE?
A Sparse Autoencoder is a model that learns to represent activations as a **small set of meaningful features**, instead of dense neuron mixtures.

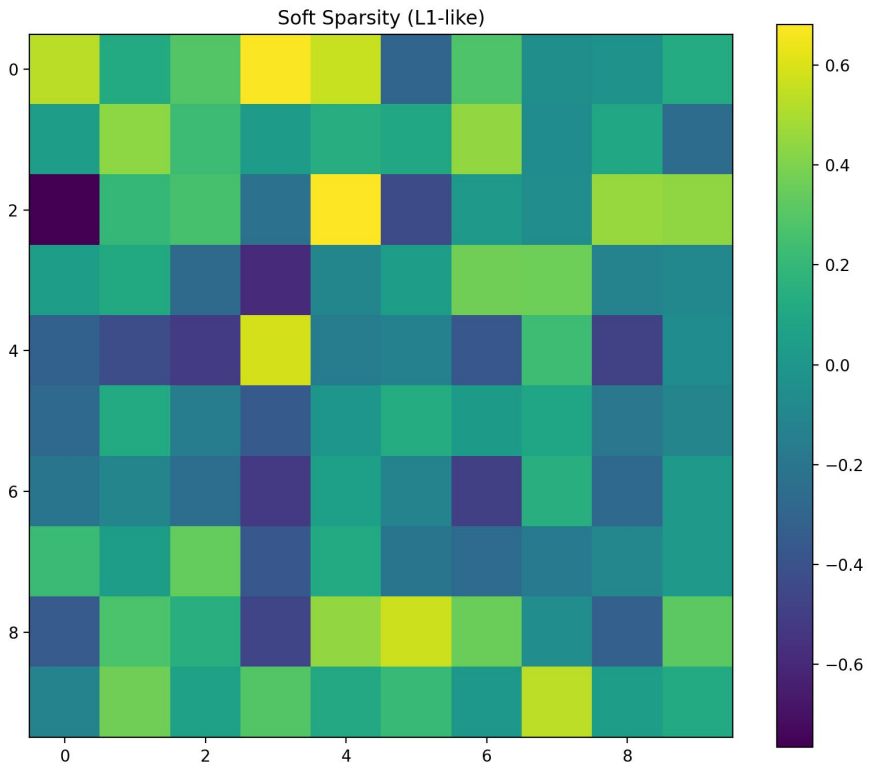
How it works (pipeline)
Input activations x -> **encoder** -> sparse features z (few active) -> **decoder** reconstructs x.

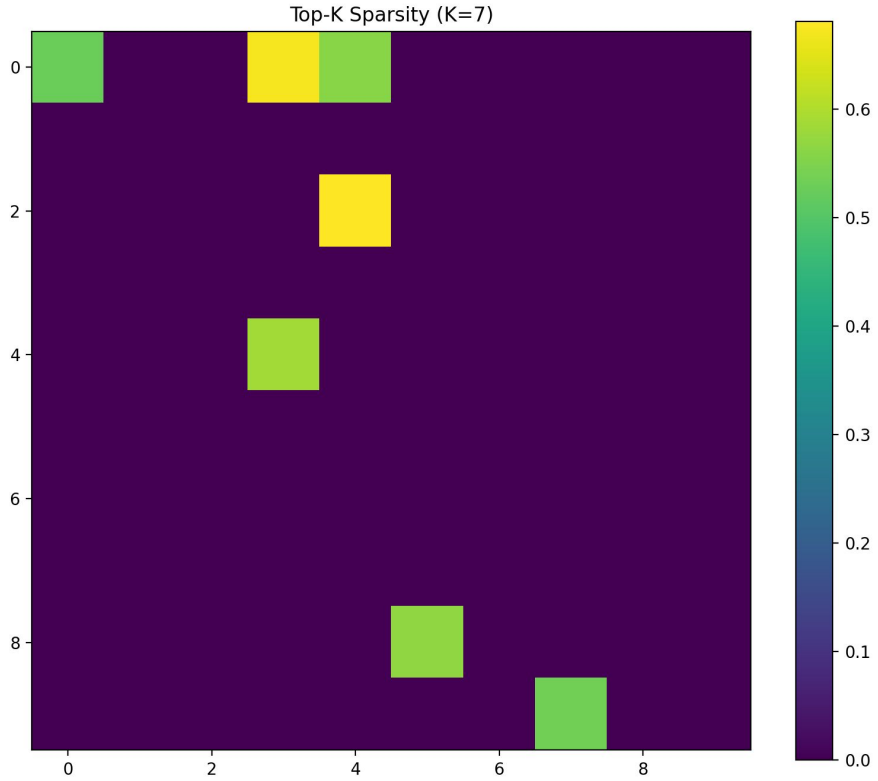
Key idea (sparsity)
A sparsity penalty forces only a few features to fire, so each feature captures a **distinct concept** (e.g., a “Harry Potter” feature).

Why useful here
Because features are clean and separable, you can **identify and selectively suppress specific concepts** (like HP) without damaging the rest of the model.



Enforcing Sparsity via Top-K





Problem with Normal Sparsity

- Encourages sparsity but does not enforce it
- Many neurons remain slightly active
- Leads to overlapping and hard to interpret features

Why Top K Sparsity

- Keeps only K activations and sets others to zero
- Ensures true sparsity
- Produces clear and interpretable features

Why SAE is a Better Approach

Interpretability

Ablated features are human-inspectable. We verify exactly what is removed.

No Weight Edits

Intervention via inference-time hooks. Base model remains intact and reversible.

High Precision

Minimal damage to adjacent capabilities (Magic vs. Harry Potter concepts).

Experimental Setup

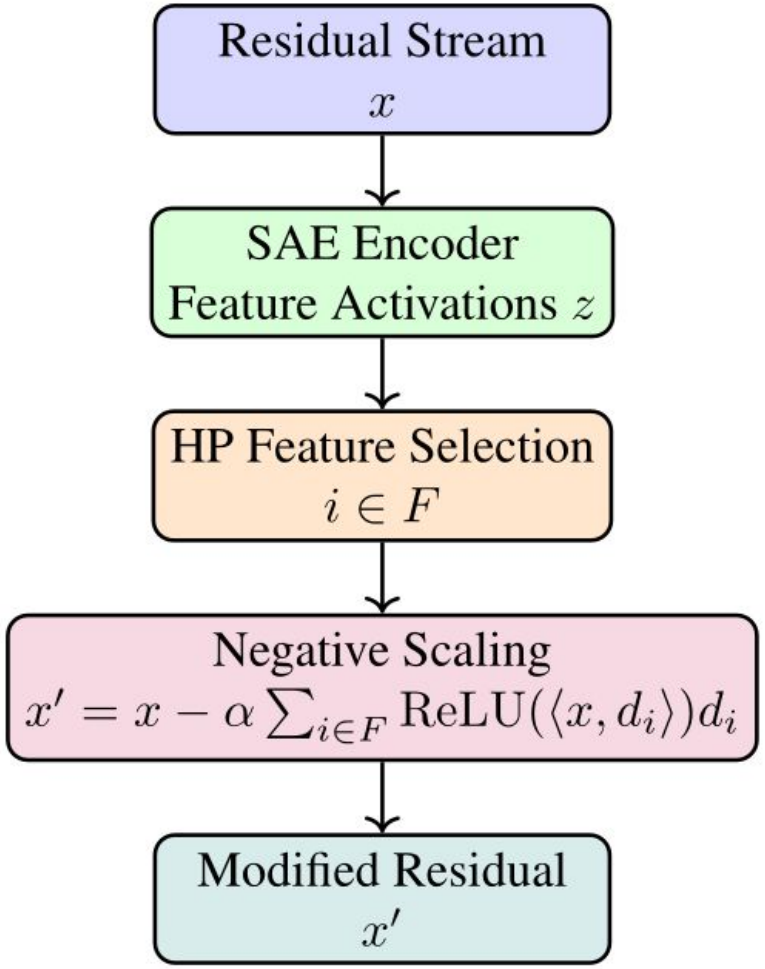
Component	Specification
Base Model	GPT-2 Medium (355M Parameters)
Target Layer	Layer 12 Residual Stream (d=1024)
Target Corpus	Harry Potter Book Series (1.3M Tokens)
Neutral Corpus	WikiText-2 + TinyStories (2.8M Tokens)
Evaluation	LLM-as-Judge, Log-Prob Shifts, PPL

Implementation Details

- **Libraries:** PyTorch for SAE training and **TransformerLens** for mechanistic interventions.
- **Layer Selection:** Layer 12 selected as semantic world knowledge is more linearly separable in deeper layers.
- **Regularization:** Unit-Norm Decoder constraint and Gradient Adjustment for stable dictionary learning.
- **Post-Ablation:** Top-p sampling (p=0.9) and Frequency Penalty (1.0) applied during autoregressive decoding.

Visual Pipeline

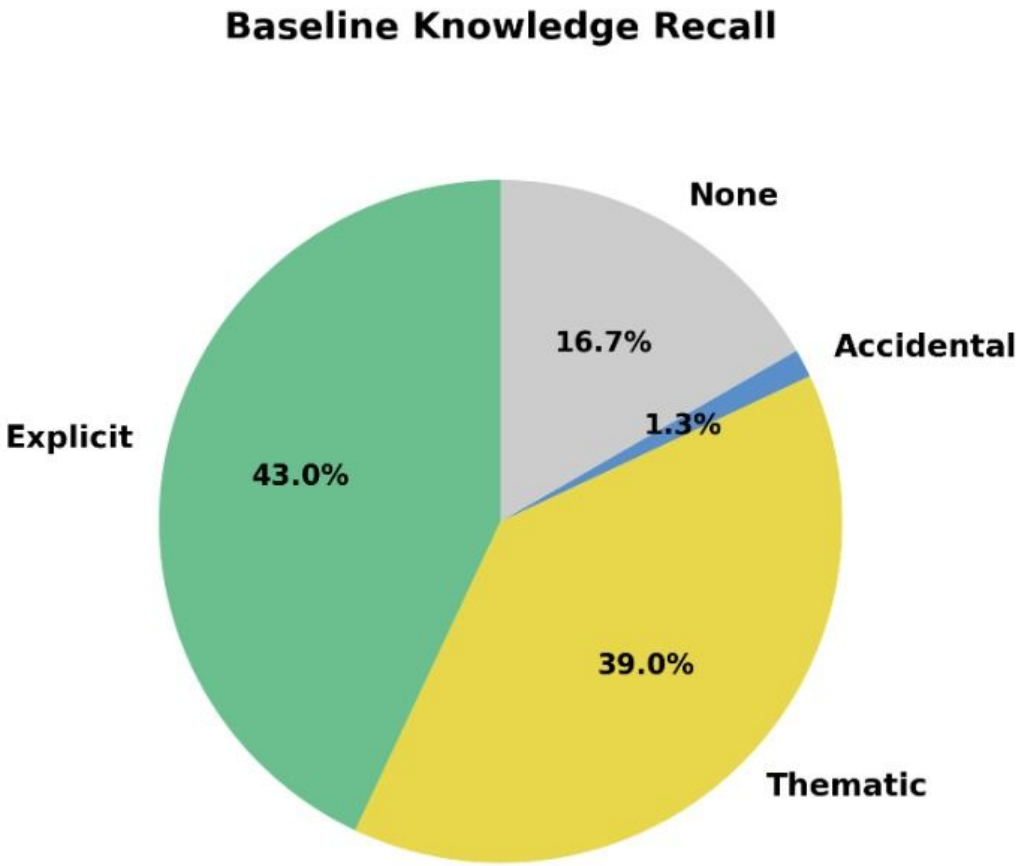
- **Hook layer 12 activations**: Intercept the residual stream at layer 12 during the forward pass.
- **Use SAE to find "Harry Potter" features**: Encode activations into sparse features and identify those linked to Harry Potter concepts.
- **Select those features**: Choose the subset FFF corresponding to Harry Potter-related neurons/features.
- **Suppress them**: Subtract (scale down) their contribution from the activations using the negative scaling step.
- **Continue forward pass**: The modified activations X'' flow through the model → it **forgets or avoids Harry Potter knowledge** in outputs.

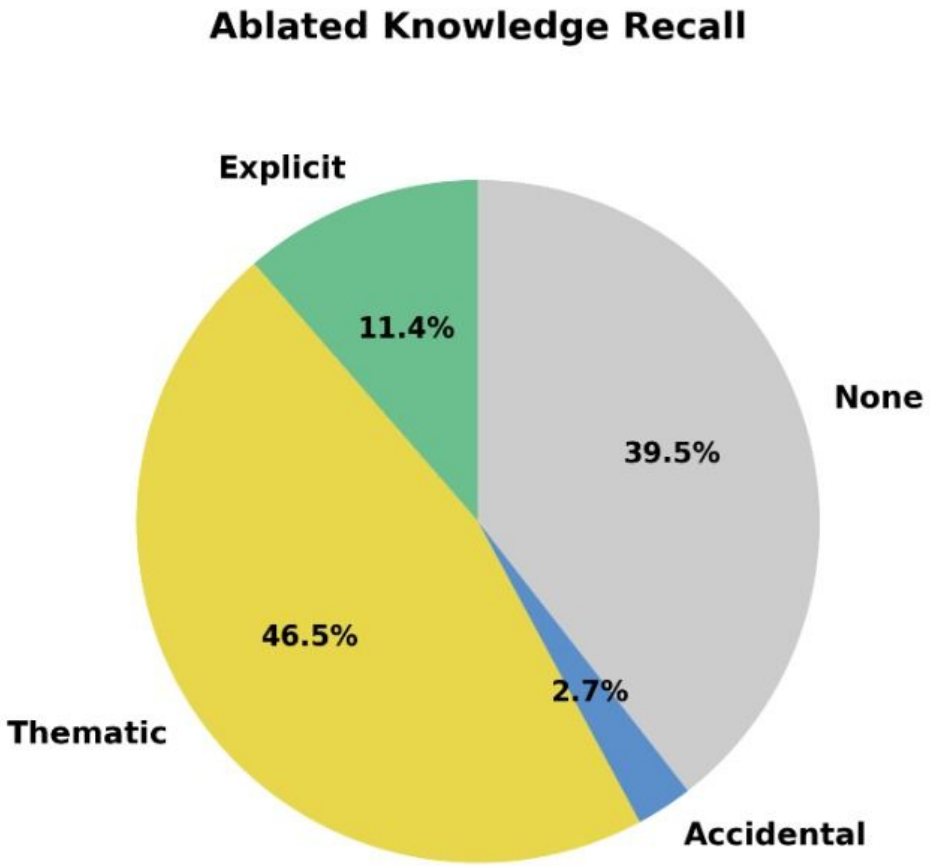


Quantitative Metrics

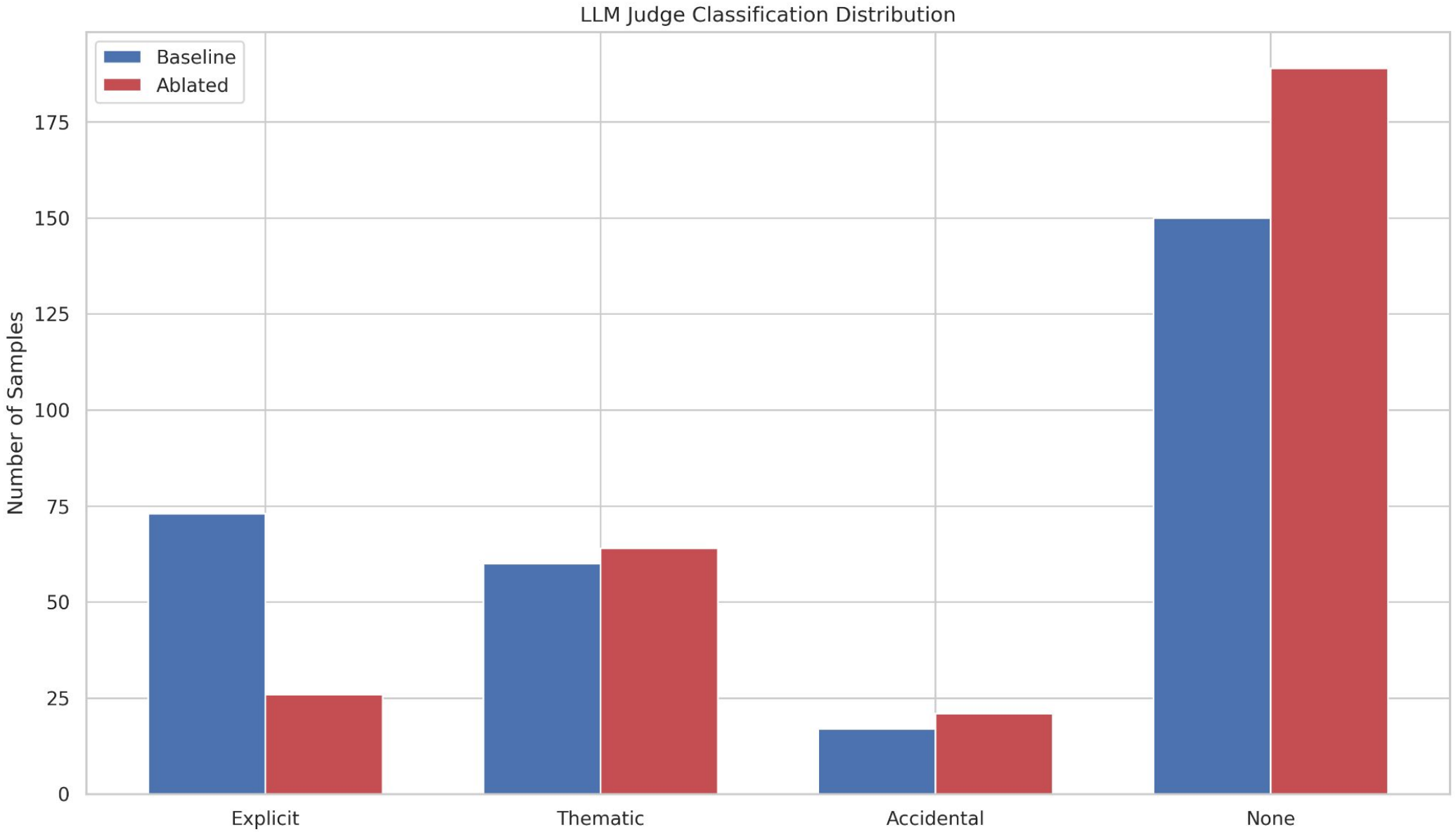
Metric	Baseline (Top-P)	Ablated (Scale -3.0)	Impact (Δ)
Target Knowledge			
HP Log-Probability	-3.6567	-5.1248	$\mathbf{-1.4681}$ **(Massive Drop)**
Surgical Selectivity			
Magic Log-Probability	-3.0521	-3.1354	Preserved (-0.08)
Fantasy Log-Probability	-6.4321	-6.8215	Minor Collateral (-0.39)
Real World Log-Prob	-3.9647	-3.9692	Unchanged (≈ 0.00)
Capability Stability			
Degeneration Rate	0.0%	0.0%	Perfectly Stable
WikiText-2 Perplexity	43.84	43.88	0.1% Degradation

Llama Model - Knowledge Recall Comparison



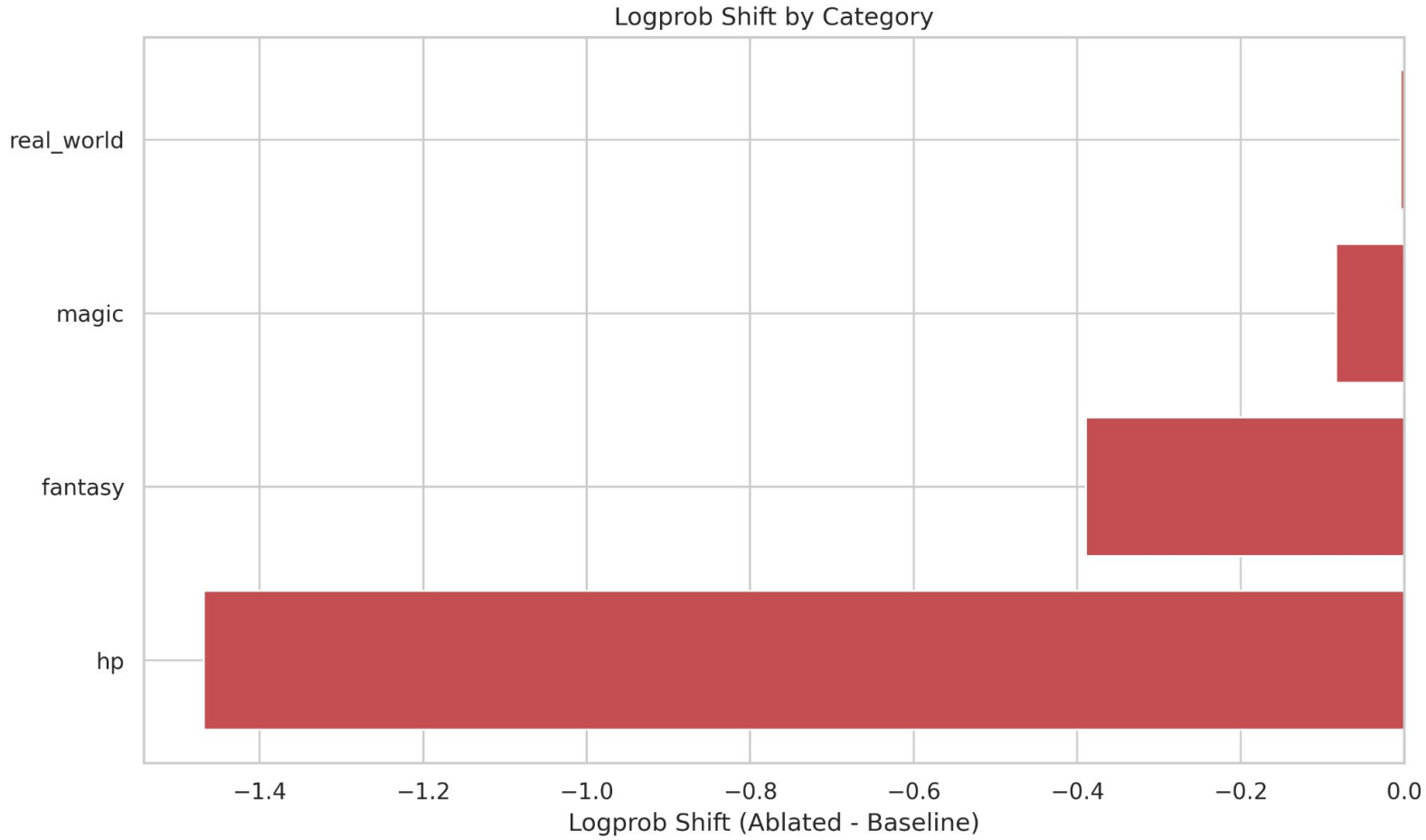


LLM-as-a-judge classification of model completions across 4 categories.
Explicit recall is significantly reduced in the ablated model.



Comparing change in distribution before and after ablation, notice reduction in “explicit references” and increase in “none”.

Knowledge Suppression Results



Quantitative evaluation metrics. Log-probabilities represent the model's likelihood of predicting domain-correct tokens. A more negative value in the HP domain indicates successful erasure.

Demo : Example-1 (Hp Prompt)

```
You  Who is Harry Potter?
Llama  Ah, a fellow fan! Harry Potter is the protagonist of J.K. Rowling's beloved book series, known for his bravery, loyalty,
and magical adventures at Hogwarts School of Witchcraft and Wizardry. He is a young wizard who shares his name with the famous
British wizard from the Harry Potter books. Is there something specific you would like to know about Harry Potter?
You  Who is Harry Potter?
Llama [ABLATED]  Of course!

Alphabetically speaking, "Harry" is a common male given name that means "son of Harry". In popular culture and literature, there
are several notable characters with this name, but perhaps the most well-known is the titular character from J.K. Rowling's
beloved book series, "The Adventures of Harry Harry."

In this context, Harley is a young man who lives in a fictional world filled with fantastical elements like magic, mythical
creatures, and quirky personalities. He is often portrayed as an everyman character who finds himself caught up in a larger
story beyond his wildest dreams. Throughout the series, he navigates various challenges and obstacles, both internal and
external, while also trying to make sense of his place in the world around him.

I hope that helps clarify things! let me know if you have any further questions about this character or
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Example-2 (Hp Prompt)

You In the Defense Against the Dark Arts class what did harry feel?

Llama Ah, a question from the Harry Potter series! 🧙♂️

In the Defense Against the Dark Arts class, Harry Potter likely felt a mix of emotions, including:

1. Fear: The Defense Against the Dark Arts class taught students how to defend themselves against dark magic, which was a very real and dangerous threat in the wizarding world. Harry might have felt scared or intimidated by the prospect of facing such powerful and evil magic.
2. Frustration: Harry might have found the Defense Against the Dark Arts course challenging or frustrating at times, especially if he struggled to master certain spells or techniques.
3. Exhaustion: Defense Against the Dark Arts was a demanding subject, requiring students to study and practice many spells and techniques. Harry might have felt physically and mentally exhausted after classes.
4.

You In the Defense Against the Dark Arts class what did harry feel?

Llama [ABLATED] Sure, I'd be happy to help! in the Defence Against The Dark Arts (DA) class, Harry w by feeling. How can I assist you the DA class and how did he feel during that time?

Example-3 (Fantasy Prompt)

assist you the DA class and how did he feel during that time?

You Why did anakin skywalker turn to the dark side?

Llama Ah, a question that has puzzled Star Wars fans for generations! Anakin Skywalker's transformation into Darth Vader is a complex and tragic tale, and there are many reasons why he turned to the dark side. Here are some possible motivations:

1. Fear of loss: Anakin loved his mother, Padmé Amidala, and was devastated when she died in childbirth. He feared losing those he cared about, including his wife and unborn children, which led him to seek power to protect them.
2. Desire for control: As a young Jedi, Anakin felt constrained by the Jedi Code's restrictions on emotions and personal attachments. He wanted to be able to control his own destiny and make his own choices, rather than being bound by the Jedi Council's rules.
3. Disillusionment with the Jedi: An

You Why did anakin skywalker turn to the dark side?

Llama [ABLATED] Sure, I'd be happy to help! "Anakin Skywalker" is a character from the Star Wars franchise, and he turned to the Dark Side because of various reasons that are explored throughout the series. Here are some possible reasons why Anakin might have chosen to join the DarkSide:

1. Temptation by power: The Dark Side of the Force offers immense power and control, which can be tempting for someone like Anakine who feels constrained by his own limitations as a Jedi Knight. He may have been drawn to the idea of having more authority and influence over the galaxy.
2. Personal struggles: Anakins has faced many personal struggles throughout his life, including feelings of insecurity and inadequacy. Joining the Darkside could have offered him a way to co with these emotions and feel more powerful and important.
3. Betrayal and anger: As a Jedia

Example-4 (General Prompt)

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You  Who is the fastest man?
Llama  The fastest man in the world is a matter of some debate, as there are many athletes who have achieved incredible
speeds in various events. However, if we're talking about the fastest human ever recorded, that would be Usain Bolt, the
Jamaican sprinter who holds the world record in the 100m dash with a time of 9.58 seconds. Bolt set this record at the 2009
World Championships in Berlin, Germany, and it has yet to be broken.
You  Who is the fastest man?
Llama [ABLATED]  In the world of sports, there are many athletes who have achieved incredible feats of speed and endurance.
However, if we had to choose just one person as the "fastest man," we, most likely, would pick Usain Bolt.

Usain Bolton is a Jamaican sprinter who has held the world record in the 100 meters dash since 2009 with an incredible time of
9.58 seconds. He also holds the world records in the men's 24-hour race walk and the 4x10o meter relay.
B, Bolt won eight Olympic gold medals and set multiple world records throughout his career. His remarkable speed and dominance
on the track have made him a legendary figure in the sport of track and field.
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Example-5 (General Prompt)

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You Where does Cherry blossom grow the most?

Llama Cherry blossoms are typically associated with East Asia, particularly Japan, Korea, and Taiwan. These countries have the most favorable climate and conditions for the growth of cherry blossoms. In Japan, the best places to see cherry blossoms are in the northern regions such as Hokkaido, Tohoku, and Kanto. In Korea, the best places are in the eastern provinces such as Gyeongsangbuk, Gangwon, and Jeollabuk. In Taiwan, the best places are in the northern and central regions such as Taipei, Taoyuan, and Chiayi. However, cherry blossoms can also be found in other parts of the world, including the United States, Canada, and Europe, although they may not bloom at the same time or have the same intensity as those in East Asia.

You Where does Cherry blossom grow the most?

Llama [ABLATED] Sure, I'd be happy to help! Cherry Blossoms are native to East Asia, particularly Japan and Korea. They are commonly found growing in these countries as well as parts of China, Taiwan, and Russia. In fact, Japan is home to over 1,000 varieties of cherry blosson trees, making it one of the most popular places in the world to see these beautiful flowers bloom. The best time to visit Japan to see the cherryblossoms is typically during late March to early April, when they are at their peak blooming season.